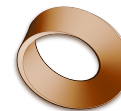


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



Refraction Investigation

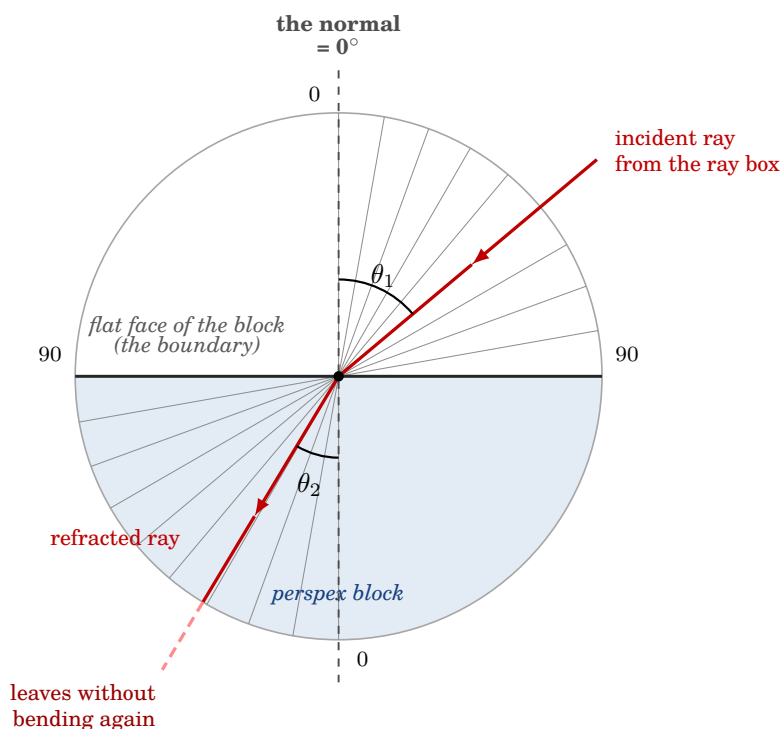
Criterion B — Inquiring and designing • Criterion C — Processing and evaluating

Learning intentions

You have already studied **reflection** — what light does when it *bounces off* a surface. This investigation is about **refraction** — what light does when it *passes into* a new material. Between them, these two are the laws that explain almost everything light does: every mirror, every lens, every pair of glasses, every camera.

Light changes direction when it crosses from one material into another. That part is easy to see. The harder question, and the one you are going to answer, is **how much** it bends — and what decides that.

Your apparatus: a ray box, a semicircular perspex block, and a printed protractor card. The block sits in the blue half of the card with its flat face along the centre line. Aim the ray at the **centre** of the card, read the **top** scale for the angle going in and the **bottom** scale for the angle coming out — and remember that on this card **the normal is 0°**.



Safety

The ray box gets **hot**. Keep paper, sleeves and pencil cases away from it, and do not touch the bulb. If you smell burning, switch it off at the power supply and tell Mr. Rawson. You will write this in your own words in the **Safety** part of B3.

B1 Your research question

You already know that light bends when it goes into a new material. So that is not the question. The question is **how much** it bends — and what you can do to change that.

Start by listing everything you can think of that might change **how much** the light bends. Write down every idea — there are no wrong ones at this stage.

One of those ideas is the easiest of all to change, because the protractor card is built for it: you can simply **swing the ray box round** and the light arrives at a different angle. Pick that one, and turn it into your research question.

Write it like this

How does changing _____ affect _____ ?
what you change *what you measure*

The first blank is your **independent variable** — the thing you change. The second is your **dependent variable** — the thing you measure.

Write your research question here.

B2 Your hypothesis

Your hypothesis is your **answer** to the research question, written before you do the experiment.

Write it like this

As the _____ increases, the _____
will **increase** / **decrease** / **stay the same**,
because _____ .

The word 'because' is where the marks are. A prediction on its own scores in the bottom band even when it turns out to be right. The reason is what lifts it.

Write your hypothesis here.

B3 Your method

Seven things. Work through them in order and your method will be complete.

Draw your set-up

Draw the apparatus **as it will actually look** on the bench, and label every piece of equipment.

The independent variable — what you change

(i) Which values will you use?

For example

“I will use angles of incidence of 0° , 10° , 20° , 30° , 40° , 50° , 60° , 70° and 80° .”

(ii) What will you measure it with? Name the instrument and its resolution.

Resolution — the word that earns the mark

Resolution is the **smallest change an instrument can show**. On a metre ruler marked in millimetres, the resolution is 1 mm.

“I will use a _____ with a resolution of _____.”

The dependent variable — what you measure

The same two things again: what you are measuring, and what you measure it with, including its resolution.

Control variables — what you keep the same

A **control variable** is something you keep the same so the test is fair. For each one, write **three** things:

1. What it is.
2. How you will keep it the same.
3. Why it matters.

A full example, all three parts

“I will keep **the temperature** the same. The classroom temperature changes very little, so doing the whole experiment in one lesson controls it. If the temperature changed, the perspex might refract light slightly differently.”

Choose only the ones that would actually change your result. If it would make no difference, leave it out — a short list of real control variables is worth more than a long list of everything you can think of.

Repeats

Write it like this

"I will repeat each measurement **3 times** and find an **average**."

Safety

Name something that could **genuinely cause injury**, and say what you will do about it.

The shape of a safety point

"_____ could cause _____. To stay safe I will _____."

Read it back

Two checks, and they are the ones that decide the top band:

- ☐ Another Year 8 could follow my method and get the same data.
- ☐ My method would actually work.

C1 Your results

Three conventions, and they carry marks on their own:

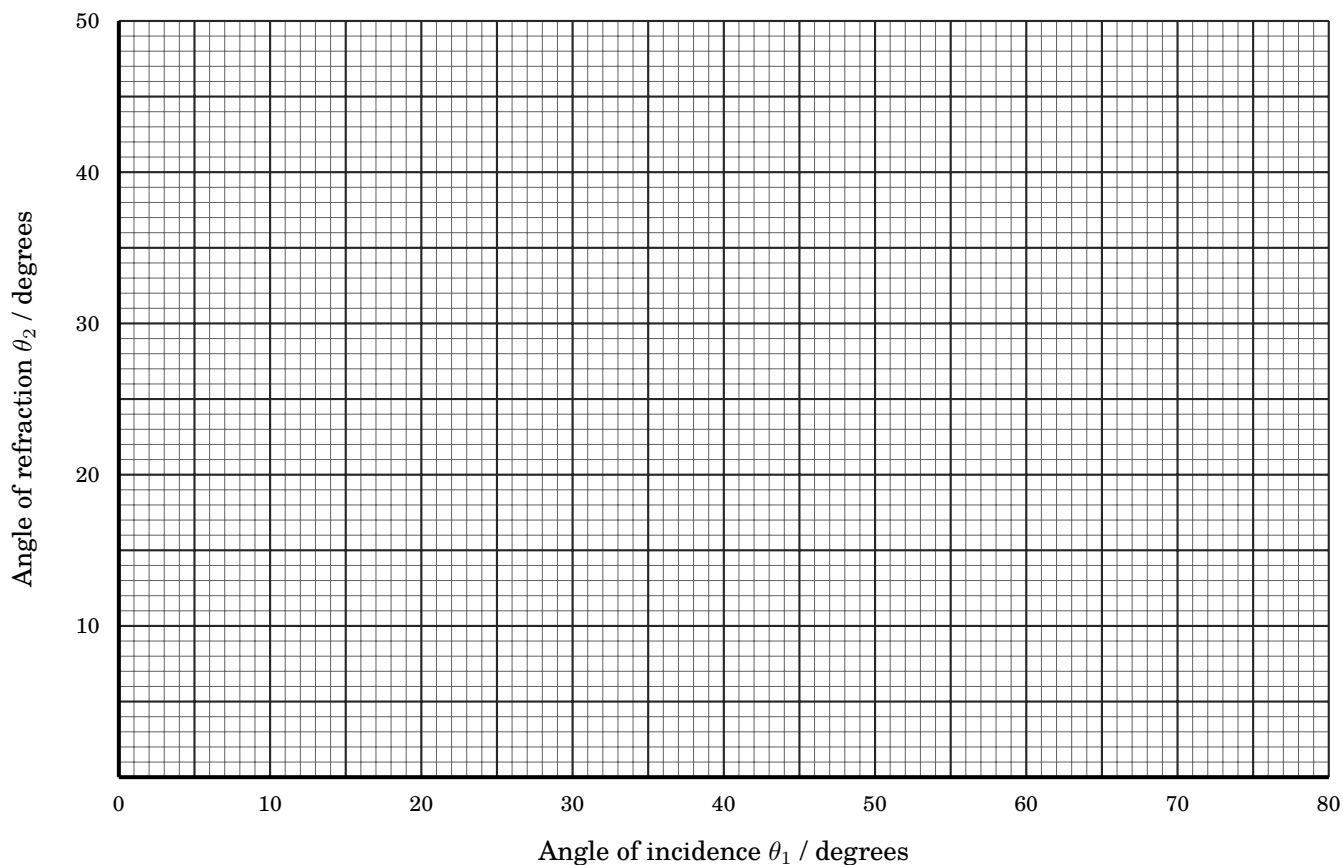
- **Units go in the heading**, not beside each number. Away from a table, the unit goes next to the number.
- The **independent variable goes in the first column**.
- Work out each **average** and write it in.

Angle of incidence θ_1 / degrees	Angle of refraction θ_2 / degrees			
	1st trial	2nd trial	3rd trial	Average
0				
10				
20				
30				
40				
50				
60				
70				
80				

Your graph

Plot the **average** angle of refraction (up) against the angle of incidence (across). Use a sharp pencil, then join your points with **one smooth curve**.

Both variables are numbers, so this is a **scatter graph with a line of best fit**. (You would only use a bar chart if the thing you changed were words rather than numbers.)



Four things the marker looks for:

- ☐ Axis numbers evenly spaced.
- ☐ Both axes labelled, with units.
- ☐ Points plotted accurately.
- ☐ A ruler used for the axes.

C2 Your conclusion

Your conclusion **answers your research question**. Look at the shape of your graph and choose the sentence that matches it.

Choose the sentence that fits your graph

Rising	As x increases, y increases.
Straight, through $(0, 0)$	x is directly proportional to y , because the graph is a straight line through the origin.
Straight, missing $(0, 0)$	There is a linear relationship between x and y , because the graph is a straight line that misses the origin.
Curving upwards	As x increases, y increases at an increasing rate.
Levelling off	However, after x is 20, y stops increasing.
No pattern	There is no correlation between x and y .

Then say **why you can say that**, in one of these three ways:

- **Numbers off your own graph** — “because when θ_1 was 20° , θ_2 was 13° , and when θ_1 was 40° , θ_2 was 26° .”
- “because the graph has a positive gradient.”
- “because the graph is curving upwards.”

Write your conclusion here.

To go further — explain it

Say **why** the physics does this. The best conclusions do not only describe the graph; they explain what is happening to the light.

C3 Was your hypothesis right?

This asks whether the prediction you made in **B2** turned out to be correct.

Write it like this

My hypothesis was **correct** / **partly correct** / **not correct**,
because _____.

Then point back at your conclusion and your data. **A hypothesis that turned out to be wrong is not a problem** — saying so clearly, and showing which data proves it, is what earns the marks here.

C4 Was your method valid?

A method is **valid** if it gave you results you can trust. Work down the left-hand column and tick everything that applies to your experiment.

My method <i>was</i> valid if...	My method was <i>not</i> valid if...
☐ The points are close to the line of best fit	☐ The points are a long way from the line of best fit
☐ A clear relationship can be seen	☐ No clear relationship can be seen
☐ Repeats were done and averaged	☐ No repeats were done
☐ The measuring instruments worked well	☐ Something went wrong with the instruments, or with doing the experiment
☐ All the variables were controlled	☐ Some variables were hard to control

Write it like this

My method was **valid** / **partly valid** / **not valid**,
because _____.

Write about the **design and the data**. Working slowly, running out of time, or a single slip are not what this question is about.

C5 Improvements and extensions

Improvements — two or more

An improvement makes the **data** better. Four questions to work through:

- (a) Could you **widen the range**? What happens at much bigger or much smaller values?
- (b) Could you take readings in **smaller steps** — going up in 5° rather than 10° ?
- (c) Was anything **hard to measure accurately**? How would you fix it?

(d) Is there a **more precise instrument** — one that would show a smaller change?

This section is about the experiment, not about how fast you worked or how neatly you wrote it up.

Extension — one new investigation

An extension asks a **new research question** with a **different** independent variable. Two things to write:

1. What the new research question would be.
2. Briefly, how you would do it.

Before you hand it in

- ☐ B1 research question names **both** variables
- ☐ B2 hypothesis has a **because**
- ☐ B3 has a labelled diagram, values, resolutions, controls, repeats and safety
- ☐ C1 table has units **in the headings** and the averages worked out
- ☐ C1 graph has both axes labelled with units
- ☐ C2 conclusion uses **numbers from your own graph**
- ☐ C3 and C4 both say **because**
- ☐ C5 has **two** improvements and **one** extension